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Zone-Reduced Six-Rule Fuzzy Speed Control Approach for Field-Oriented Induction Motor Drives

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Abstract

This paper evaluates a zone-reduced rule-base simplification approach for a Mamdani-type fuzzy speed controller used in the field-oriented control of an induction motor drive. Starting from a 25-rule fuzzy controller, the rule base is reduced first to a 9-rule structure through dominant-polarity grouping and then to a 6-rule structure through macro-zone representation of the reduced input space. The reduced controllers are evaluated under identical simulation conditions using transient-response indices, integral error criteria, MATLAB profiler-based execution-time measurements, and fuzzy c-means-based decision-surface similarity analysis. The 9-rule controller provides the most favorable performance–complexity compromise among the investigated fuzzy configurations, reducing the overshoot from 1.2348% to 0.5051% and achieving the lowest IAE, ITAE, and ISE values. The 6-rule controller provides the largest relative reduction in MATLAB profiler execution time, with a 57.77% decrease compared with the 25-rule controller. The FCM-based analysis indicates structural similarity between the original and reduced decision surfaces, with R^2 values of 0.8593 and 0.8159 for the 9-region and 6-region representations, respectively. These results suggest that the proposed simplification approach can reduce fuzzy rule-base complexity for the investigated simulation case while maintaining acceptable speed-tracking performance. However, real-time embedded implementation and experimental validation are left for future work.

Keywords: induction motor drives, field-oriented control, fuzzy logic control, zone reduction, machine-learning-based validation, rule-base simplification.

1. Introduction

Induction motors are widely used in industrial drive systems because of their robust construction, low maintenance requirements, cost effectiveness, and suitability for variable-speed operation. Saidur et al. [1] highlighted the energy-saving potential of variable-speed drives in electrical-motor applications, whereas Humza and Kim [2] addressed the importance of accurate equivalent-circuit parameter estimation in induction-motor analysis and design. Nevertheless, high-performance control of induction motors remains challenging because these machines are nonlinear, multivariable, and strongly coupled electromechanical systems. Field-oriented control (FOC) addresses this difficulty by decoupling the flux-producing and torque-producing current components. Finch and Giaouris [3] discussed the principles and implementation of controlled AC drives, while Bose [4] reviewed major advances in power-electronic motor-drive systems. El-Ongeli and colleagues [5] studied recent high-performance asynchronous motor drive control systems. By separating flux and torque, FOC gives asynchronous motors control capabilities like separately stimulated DC motors.

The performance of a field-oriented control (FOC) actuator depends largely on the controllers employed in the inner and outer control loops. Proportional-integral controllers are commonly employed because of their simple form and easy implementation. However, the performance of fixed-gain PI controllers may decline due to variations of machine parameters, model mismatch, nonlinear operating circumstances and external disturbances. Fuzzy logic control is another possibility where the required control action can be described by linguistic rules instead of a complete high-order mathematical representation of the plant. Kurt and Kiak [6] demonstrated the use of fuzzy decision processes in intelligent lighting control. Prikop and Hollander [7] evaluated a wide range of successful applications of industrial fuzzy control. Jantzen [8] presented the theoretical foundations for the design and implementation of a fuzzy controller. In electric drive systems, Khomyan et al. [9] developed a fuzzy speed controller for a separately excited DC motor, and Mendoza-Mondragon et al. [10] investigated robust speed control for permanent magnet synchronous motors. Both of these studies were conducted in the context of electric drive systems. Specifically, El-Barbari [11] applied fuzzy logic to a five-phase induction motor drive, Manan et al. [12] took into account the effects of core losses in fuzzy speed control, and Zeb et al. [13] proposed an adaptive fuzzy controller for an induction motor drive with indirect current control. All of these studies were conducted for induction motor applications. Although the controller structure and design objectives in these studies are different from the rule-based simplification problem considered in this study, these works confirm the suitability of fuzzy control for solving nonlinear speed control problems. This is a practical limitation for fuzzy controllers because their computational overhead often increases with the increase in the number of input variables, linguistic terms, and inference rules. This is another limitation of fuzzy controllers. The larger set of rules in a Mamdani-type fuzzy inference system leads to an increase in the number of operations required for fuzzification, rule activation, aggregation, and defuzzification of the center of gravity. This is particularly relevant for motor control applications, where the controller must be regularly evaluated under continuous sampling conditions. This problem was directly addressed by Taleb et al. [14], who used a simplified fuzzy speed controller to drive an asynchronous motor and showed that the number of rules can be reduced while maintaining acceptable dynamic behavior. Taking a more general rule-based design perspective, Gacto et al. [15] showed that compactness, interpretability, and redundancy are closely related in linguistic fuzzy systems. Chan and Shi [16] also studied the design and development of intelligent devices for induction motor drives. But there is no explanation in the current literature on how the initial uncertainty structure is maintained when its rules are decreased. Other works have concentrated on the optimization of the controller and not the rules, through the use of structured zone fusion. Ali et al. [17] proposed fuzzy speed controller for induction motor by using backtracking search algorithm. Bachache and Wen [18] used particle swarm optimization to design fuzzy membership functions for AC drive, while Farah et al. [19] investigated the effectiveness of the network to improve the efficiency of the induction motor. Mugheri and Keerio [20] designed an optimized PI controller for induction motor operating under V/f scalar control. In a more broad way, from a theoretical point of view, Fazzolari et al. [21] studied many optimization factors and illustrated how precision, robustness and interpretability may be obtained at the same time in the design of controllers. This optimization technique can be utilized to optimize member engagement, new functions, administrator allocation or regulatory burdens. Nevertheless, they do not directly address the problem considered here: systematically reorganizing an existing fixed Mamdani rule table into progressively smaller, coverage-preserving decision structures without independently retuning the reduced controllers.

Recent fuzzy-control research has expanded toward learning-assisted, event-triggered, networked, and security-aware systems. Yan et al. [22] combined deep Q-learning with proportional–integral event-triggered fuzzy output control for variable-speed automated guided vehicles. Yan and Yang [23] proposed neural-network-based attack-compensation control for T–S fuzzy systems subject to actuator attacks and introduced an improved dynamic memory event-triggered mechanism. Yan et al. [24] studied synchronization of delayed fuzzy neural networks under probabilistic communication delays and demonstrated its application to image encryption. In a safety-critical medical application, Yan et al. [25]

developed memory-based attack-tolerant T–S fuzzy control for a networked artificial-pancreas system exposed to false-data-injection attacks. These studies address learning-based adaptation, communication efficiency, network synchronization, and cyberattack tolerance, respectively, rather than structural rule-base simplification.

The applicability of advanced fuzzy control has also been demonstrated in energy, robotic, and electric-drive systems. Yan et al. [26] proposed sampled memory-event-triggered fuzzy load-frequency control for wind-power systems affected by outliers and transmission delays. Precup et al. [27] investigated fuzzy control for telesurgical systems, in which delay, precision, constrained motion, and human–robot interaction are central design concerns. Hu et al. [28] developed a dynamic prescribed-performance fuzzy-neural backstepping controller for a permanent-magnet synchronous motor under step-load conditions. Andonovski et al. [29] surveyed recent data-driven evolving fuzzy and neuro-fuzzy control methods and emphasized trends toward online adaptation, model evolution, and formal control analysis. Mao et al. [30] investigated adaptive fuzzy tracking control with global prescribed-time prescribed performance for uncertain nonlinear systems. Although these studies demonstrate the broad capabilities of contemporary fuzzy control, they generally introduce adaptive, neural, event-triggered, networked, or performance-constrained mechanisms. The present work instead addresses a narrower but practically relevant question: whether the structural complexity of a fixed Mamdani fuzzy speed controller can be reduced systematically while retaining its dominant decision organization and acceptable closed-loop behavior under controlled simulation conditions. Accordingly, the objective of this study is to develop and evaluate a structured zone-reduction method for the Mamdani fuzzy speed controller of an FOC-based induction-motor drive. The novelty does not reside in fuzzy rule reduction alone, which has already been investigated, but in the combination of a reproducible two-stage reduction procedure and a multilayer validation framework. The main contributions are as follows. First, the original 25-rule, 5×5 fuzzy decision matrix is reduced to a 9-rule, 3×3 structure through an explicit mapping of the original linguistic terms into negative, equilibrium, and positive polarity classes. Second, the 9-rule structure is further represented by six macro-rules using neighborhood continuity, dominant output polarity, compensation behavior, and coverage constraints rather than arbitrary rule deletion. Third, every cell of the reduced 3×3 linguistic input space remains assigned to a representative macro-rule, thereby preserving full reduced-space coverage. Fourth, the membership-function shapes, scaling factors, universes of discourse, inference mechanism, aggregation procedure, sampling conditions, and centroid-defuzzification method are kept unchanged across the 25-rule, 9-rule, and 6-rule fuzzy controllers, so that the primary comparison isolates the effect of rule-base restructuring. Finally, the reduced controllers are evaluated using several complementary criteria. Orłowska-Kowalska et al. [31] provide the broader drive-control context for performance assessment, while Workineh et al. [32] demonstrate the use of cumulative error indices in induction-motor speed-control evaluation. For structural analysis, the FCM formulation follows the scalable clustering framework discussed by Havens et al. [33], while cluster-validity considerations are supported by Wang et al. [34] and numerical aspects of fuzzy-clustering validation are considered in light of Eustaquio and Nogueira [35]. Accordingly, this study combines transient-response indices, IAE, ITAE, and ISE, MATLAB profiler-based execution-time measurements, and FCM-based decision-surface analysis. The 9-rule controller is examined as a balanced performance–complexity configuration, whereas the 6-rule controller is evaluated as a more aggressively simplified alternative.

The remainder of this paper is organized as follows. Section 2 presents the induction-motor model and the FOC structure. Section 3 describes the baseline fuzzy controller and the proposed two-stage zone-reduction procedure. Section 4 presents and discusses the transient-response, integral-error, profiler-based execution-time, and FCM-based structural-validation results. Section 5 summarizes the main findings, limitations, and future research directions.

2. Methodology

In order to realize high-performance control, the induction motor is modeled in the synchronous rotating dq reference frame. This representation transforms the three-phase nonlinear system into a form more suitable for control design by expressing stator and rotor variables in orthogonal direct-axis and quadrature-axis components. The synchronous reference-frame model is particularly convenient for field-oriented control because it enables the decoupled representation of flux and torque dynamics.

The equations for the stator voltage are given by

$$v_{ds} = R_s i_{ds} + \frac{d\psi_{ds}}{dt} - \omega_e \psi_{qs} \quad (1)$$

$$v_{qs} = R_s i_{qs} + \frac{d\psi_{qs}}{dt} - \omega_e \psi_{ds} \quad (2)$$

The equations for rotor voltage are given by

$$v_{dr} = R_r i_{dr} + \frac{d\psi_{dr}}{dt} - (\omega_e - \omega_r) \psi_{qr} \quad (3)$$

$$v_{qr} = R_r i_{qr} + \frac{d\psi_{qr}}{dt} - (\omega_e - \omega_r) \psi_{dr} \quad (4)$$

where R_s and R_r denote stator and rotor resistances, i denotes current, ψ denotes flux linkage, ω_e is the synchronous electrical angular speed, and ω_r is the rotor electrical angular speed. For the squirrel-cage induction motor considered in this study, the rotor windings are short-circuited; hence

$$v_{dr} = 0, \quad v_{qr} = 0 \quad (5)$$

The flux-linkage equations are expressed as

$$\psi_{ds} = L_s i_{ds} + L_m i_{dr} \quad (6)$$

$$\psi_{qs} = L_s i_{qs} + L_m i_{qr} \quad (7)$$

$$\psi_{dr} = L_r i_{dr} + L_m i_{ds} \quad (8)$$

$$\psi_{qr} = L_r i_{qr} + L_m i_{qs} \quad (9)$$

where L_s , L_r , and L_m are the stator, rotor, and mutual inductances, respectively. The electromagnetic torque is given by

$$T_e = \frac{3p}{2} (\psi_{ds} i_{qs} - \psi_{qs} i_{ds}) \quad (10)$$

where p is the number of poles. The mechanical dynamics of the rotor are defined by

$$J \frac{d\omega_m}{dt} = T_e - T_L - B\omega_m \quad (11)$$

where J is the moment of inertia, T_L is the load torque, B is the viscous friction coefficient, and ω_m is the mechanical rotor speed. These equations constitute the mathematical foundation of the induction motor model used throughout the study.

The principle of field-oriented control is to decouple the control of torque and flux by aligning the rotor-flux vector with one axis of the synchronous rotating reference frame. In this study, rotor-flux orientation is adopted, and the rotor flux is aligned with the d-axis. Under this condition,

$$\psi_{qr} = 0, \quad \psi_{dr} = \psi_r \quad (12)$$

As a result, the flux-producing and torque-producing components become decoupled. The d-axis current i_{ds} primarily determines the rotor flux, whereas the q-axis current i_{qs} determines the electromagnetic torque. Under rotor-flux orientation, the torque equation becomes

$$T_e = \frac{3}{2} \frac{p}{2} \left(\frac{L_m}{L_r} \right) \psi_r i_{qs} \quad (13)$$

The rotor time constant is defined as

$$\tau_r = \left(\frac{L_r}{R_r} \right) \quad (14)$$

and the rotor flux dynamics can be approximated by

$$\psi_r = \frac{L_m}{(1 + \tau_{rs})} i_{ds} \quad (15)$$

The slip angular speed is obtained as

$$\omega_{sl} = \frac{R_r L_m}{L_r \psi_r} i_{qs} \quad (16)$$

and the synchronous electrical speed is determined by

$$\omega_e = \omega_r + \omega_{sl} \quad (17)$$

Thus, by regulating i_{ds} and i_{qs} , the induction motor can be controlled in a decoupled manner, similar to a separately excited DC motor.

Parameter	Value	Unit
Rated voltage	690	V
Pole pairs	2	-
Stator resistance	2.75e-3	Ω
Rotor resistance	2.75e-3	Ω
Stator leakage inductance	0.089e-3	H
Rotor leakage inductance	0.089e-3	H
Mutual inductance	2.6e-3	H
Moment of inertia	0.3	kg*m ²
Viscous friction coefficient	1.1e-3	N*m*s/rad

Table 1. Induction motor parameters used in simulation.

The induction motor and drive-system parameters used throughout the simulations are listed in Table 1. All controller configurations were evaluated using the same machine constants, mechanical parameters, inverter settings, and sampling conditions in order to ensure a consistent and fair comparison.

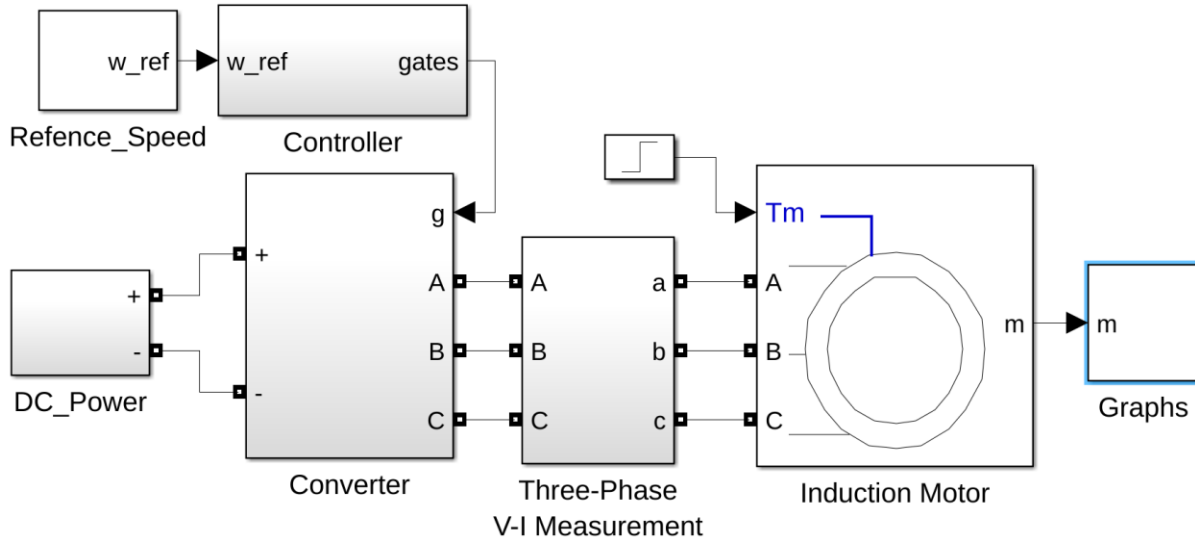


Figure 1. Overall field-oriented control structure of the induction motor drive system.

The overall control structure used in this study consists of an outer speed-control loop and an inner vector-control approach as shown in figure 1. The speed reference ω^* is compared with the measured rotor speed ω_m , and the resulting speed error is processed by the controller to generate the torque-producing current reference i_{qs}^* . The flux-producing current reference i_{as}^* is selected according to the desired operating flux level.

The three-phase stator currents are transformed into the stationary α - β frame by Clarke transformation and then into the synchronous d-q frame by Park transformation. After the controller generates the required control action, the inverse transformations are used to reconstruct the three-phase voltage commands for the inverter. In this way, the FOC structure ensures independent regulation of torque and flux channels.

3. Controller Design

The baseline controller used in this study is a Mamdani-type fuzzy logic controller embedded in the speed loop of the FOC system. It is designed as a two-input, one-output inference system. The two inputs are the speed error $e(k)$ and the change in speed error $\Delta e(k)$, whereas the output corresponds to the control action that determines the torque-producing response of the drive.

Each rule in the fuzzy system has the general form:

$$\text{IF } e \text{ is } A_i \text{ AND } \Delta e \text{ is } B_j \text{ THEN } u \text{ is } C_{ij} \quad (18)$$

where A_i and B_j denote the linguistic labels for input variables and C_{ij} denotes the output linguistic term. The fuzzy inference process consists of fuzzification, rule evaluation, aggregation, and defuzzification. Centroid defuzzification is used to obtain the crisp controller output. To ensure that the comparison isolates the effect of rule-base simplification, the membership-function shapes, input and output universes of discourse, scaling factors, Mamdani inference mechanism, aggregation procedure, sampling conditions, and centroid defuzzification settings were kept unchanged for the 25-rule, 9-rule, and 6-rule controllers.

Therefore, the reduced controllers were not independently retuned controllers; only the rule-base structure was modified.

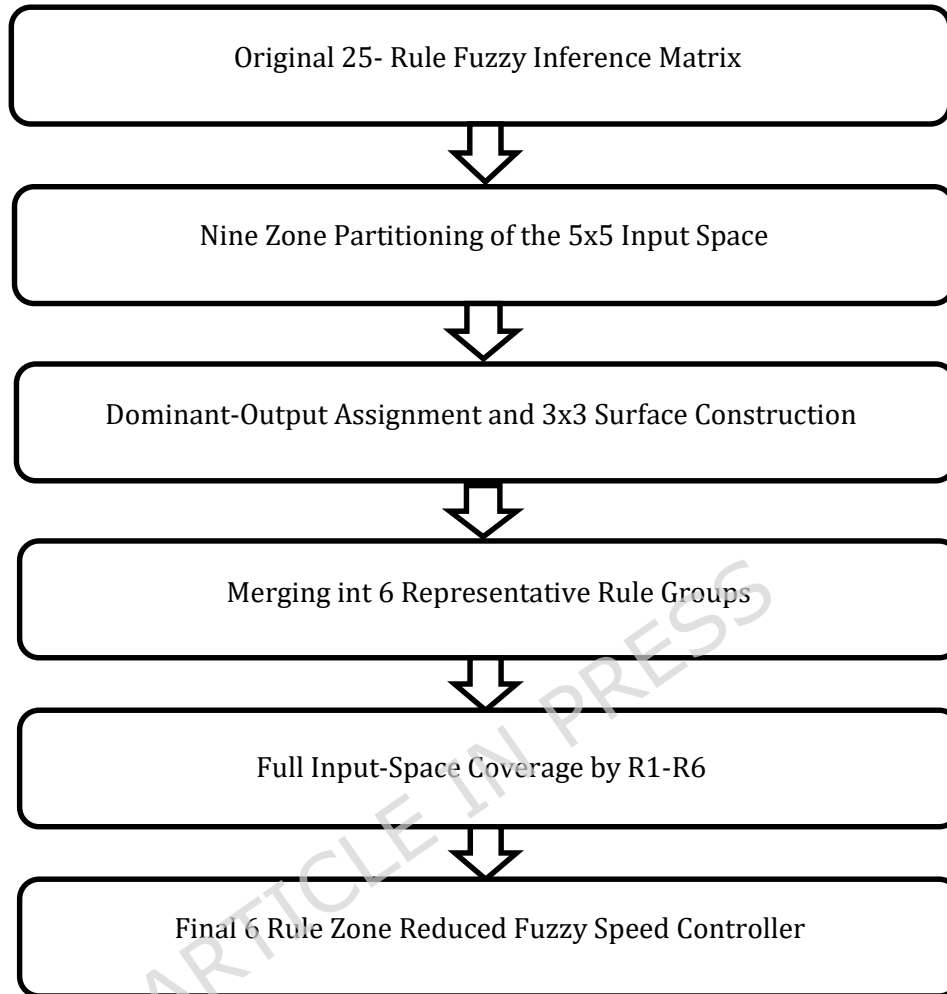


Figure 2. Proposed zone-reduction workflow from the original 25-rule fuzzy inference matrix to the final six-rule fuzzy speed controller.

Figure 2 illustrates the proposed two-stage zone-reduction approach used to simplify the original fuzzy inference system from 25 rules to 9 rules and finally to 6 rules. In the first stage, the original 5×5 fuzzy inference matrix is compressed into a reduced 3×3 rule matrix by grouping the output linguistic terms according to dominant negative, equilibrium, and positive control tendencies. This operation preserves the dominant polarity structure of the original decision surface while reducing the rule count. In the second stage, the intermediate 3×3 structure is repartitioned according to a neighborhood-preserving zone-reduction logic. Adjacent cells with similar control tendencies are grouped into macro-regions so that the final 6-rule controller retains the dominant negative, equilibrium, and positive decision behavior of the original fuzzy controller with a lower implementation cost.

Table 2 presents the original 25-rule decision matrix used as the reference controller. The table indicates the expected monotonic structure: strongly negative errors are associated with negative control actions, strongly positive errors are associated with positive actions, and the central region around the equilibrium

point is controlled by the zero-control rule. This matrix therefore defines the original nonlinear decision surface that is later simplified through zone-based reduction.

Ce/E	NB	NS	Z	PS	PB
NB	NB	NS	Z	NB	NB
NS	NS	NS	Z	NB	NS
Z	Z	NS	Z (0.01)	PB	PB
PS	PS (0.01)	PS (0.01)	Z	PS (0.01)	PS (0.01)
PB	PS (0.01)	PS (0.01)	Z	PS (0.01)	PS (0.01)

Table 2. Original 25-rule fuzzy inference matrix used in the baseline Mamdani controller.

The original 25-rule matrix is reduced in the first stage by merging adjacent output linguistic terms according to dominant polarity. Specifically, the negative labels {NB, NS} are mapped to N, the central label Z is preserved, and the positive labels {PS, PB} are mapped to P. First the rules will be divided to 9 zones so after that can be simplified to smaller groups based on the naturality of the zone as can be seen in the table below:

Ce/E	NB	NS	Z	PS	PB
NB	Zone 1		Zone 2	Zone 3	
NS					
Z	Zone 4		Zone 5	Zone 6	
PS	Zone 7		Zone 8	Zone 9	
PB					

Table 3. Nine-zone partitioning of the original 5×5 fuzzy input space for first-stage rule-base reduction.

Table 3 illustrates the first-stage partitioning of the original 5×5 fuzzy input space into nine macro zones. In this structure, the input linguistic labels NB and NS are grouped into the negative region, Z is preserved as the equilibrium region, and PS and PB are grouped into the positive region. Each macro zone (presenting in table 4) represents a neighboring block of rules with similar input characteristics and serves as an intermediate grouping step before constructing the reduced 3×3 fuzzy rule matrix. Thus, table 3 shows how the original 25-rule structure is systematically reorganized into nine dominant decision regions.

Based on the nine-zone partitioning shown in table 3, each macro zone is evaluated in terms of the output labels inherited from the original 5×5 fuzzy rule matrix. The representative output of each macro zone is then selected according to the dominant polarity criterion. Table 4 summarizes the covered output labels, the selected dominant output, and the interpretation of each macro zone. This step forms the intermediate basis for constructing the reduced 3×3 fuzzy rule matrix in the next stage.

The terms dominant negative, equilibrium, and positive control tendency are defined through an explicit output-polarity mapping. Let the original output linguistic set be $Y = \{NB, NS, Z, PS, PB\}$.

The reduced polarity mapping $\pi(y)$ is defined as:

$$\pi(y) = N, \text{ if } y \in \{NB, NS\} \quad 19$$

$$\pi(y) = Z, \text{ if } y \in \{Z\} \quad 20$$

$$\pi(y) = P, \text{ if } y \in \{PS, PB\} \quad 21$$

Accordingly, the terms negative, equilibrium, and positive do not denote qualitative interpretations only; they correspond to the reduced output-polarity classes N, Z, and P obtained from the original fuzzy output labels.

Macro zone	Covered output labels in 5×5 map	Dominant output	Interpretation
Zone 1	NB, NS, NS, NS	N	Strong negative region
Zone 2	Z, Z	N	Moderate negative region
Zone 3	NB, NB, NB, NS	N	Transition balancing region
Zone 4	Z, NS	Z	Equilibrium/compensation region
Zone 5	Z (0.01)	Z	Central equilibrium region
Zone 6	PB, PB	P	Strong positive region
Zone 7	PS (0.01), PS (0.01), PS (0.01), PS (0.01)	P	Weak positive region
Zone 8	Z, Z	Z	Equilibrium region
Zone 9	PS (0.01), PS (0.01), PS (0.01), PS (0.01)	P	Weak positive region

Table 4. Dominant output assignment of the nine macro zones obtained from the original 5×5 fuzzy rule matrix.

As shown in table 4, the original 5×5 fuzzy rule matrix is systematically reorganized into nine macro zones according to the dominant polarity characteristics of the input variables. Each macro zone groups neighboring fuzzy rules that exhibit similar control behavior. For macro zones containing multiple output labels, the representative output is determined by the most frequently occurring dominant polarity within that region. In macro zones containing identical or single dominant outputs, the representative output is directly assigned without additional majority evaluation.

This dominant output assignment keeps the main negative, equilibrium, and positive control tendencies of the original rule surface while reducing the structural complexity of the inference mechanism. After this stage, the nine macro zones are mapped onto a compact 3×3 fuzzy decision surface.

The original fuzzy controller employs five linguistic variables for both the error (e) and change of error (ce) which are $\{NB, NS, Z, PS, PB\}$.

where:

NB: Negative Big

NS: Negative Small

Z: Zero

PS: Positive Small

PB: Positive Big

The original fuzzy rule base can be represented as:

$$R = [r_{ij}]_{5 \times 5} \quad (22)$$

where each element r_{ij} denotes the fuzzy output corresponding to the pair (e_i, ce_j) .

To simplify the inference mechanism, the linguistic variables were grouped into three dominant polarity classes:

$$N = \{NS, NB\}$$

$$Z = \{Z\}$$

$$P = \{PS, PB\}$$

Accordingly, a mapping function $g(x)$ was defined as:

$$g(x) = \begin{cases} N, & x \in \{NS, NB\} \\ Z, & x = Z \\ P, & x \in \{PS, PB\} \end{cases} \quad (193)$$

This transformation compresses the original 5x5 rule matrix into nine macro zones:

$$Z_k = \{r_{ij} : g(e_i) = a, g(ce_j) = b\} \quad (24)$$

where:

$$a, b \in \{N, Z, P\}$$

Each macro zone contains a subset of the original rules sharing similar polarity characteristics.

To determine the representative output of each macro zone, a dominant output selection mechanism based on majority occurrence was applied. The dominant output D_k for the k^{th} macro zone is obtained by:

$$D_k = \arg \max_{y \in Y} n_k(y) \quad (205)$$

where:

$$Y = \{NB, NS, Z, PS, PB\}$$

and:

$$n_k(y) = \sum_{r_{ij} \in Z_k} I(r_{ij} = y) \quad (216)$$

The indicator function $I(\cdot)$ is defined as:

$$I(r_{ij} = y) = \begin{cases} 1, & r_{ij} = y \\ 0, & r_{ij} \neq y \end{cases} \quad (227)$$

Thus, the output label with the highest occurrence frequency inside the macro zone is selected as the dominant representative.

Finally, the selected dominant outputs are also compressed into the reduced polarity set $\{N, Z, P\}$ using:

$$h(y) = \begin{cases} N, & y \in \{NB, NS\} \\ Z, & y = Z \\ P, & y \in \{PS, PB\} \end{cases} \quad (238)$$

Hence, the reduced fuzzy inference matrix is expressed as:

$$R_{red}(a, b) = h \left(\arg \max_{y \in Y} n_{ab}(y) \right) \quad (249)$$

leading to the final reduced 3x3 fuzzy decision surface:

$$R_{red} = \begin{bmatrix} O_1 & O_2 & O_3 \\ O_4 & O_5 & O_6 \\ O_7 & O_8 & O_9 \end{bmatrix} \quad (25)$$

After dominant-polarity reduction, table 4 shows the reduced 9-rule matrix. This intermediate controller has two functions. First, it simplifies but faithfully represents the fuzzy surface. Second, it establishes a logical starting point for the second reduction step and final 6-rule FIS.

The final six-rule fuzzy inference system was derived from a decreased 3x3 fuzzy decision surface using a second-stage zone-reduction approach. Nine reduced decision rules were not treated independently at this level. Common macro rule groups were formed from nearby cells with similar dominant control tendencies. This method maintains the key negative, equilibrium, and positive control actions of the 3x3 decision surface while decreasing the number of examined fuzzy rules from nine to six. Table 5 lists the final six typical rules, their reduced cells, outputs, and physical control interpretations.

Rule Groups	Covered reduced cells	Output	Control role
R1	(N, N)	N	Strong negative correction for large negative error tendency
R2	(N, P)	N	Negative transition balancing during opposite-sign error variation
R3	(N, Z), (Z, N)	Z	Near-equilibrium damping and compensation on the negative side
R4	(Z, Z), (P, Z)	Z	Central equilibrium and positive-side zero-transition coverage
R5	(Z, P)	P	Positive transition correction
R6	(P, N), (P, P)	P	Positive compensation and strong positive correction

Table 5. Second-stage 9-to-6 macro zone merging for the six-rule fuzzy inference system.

As shown in Table 5, Rules R1 and R2 represent the dominant negative-control regions of the reduced decision surface. Rules R3 and R4 describe the near-equilibrium and compensation regions, where the controller output is assigned to the zero-control action in order to avoid excessive correction around the reference speed. Rule R5 represents the positive transition region, while Rule R6 groups the positive-side compensation and strong positive-control regions. Therefore, the final six-rule structure does not discard any part of the reduced input space; instead, it represents multiple neighboring cells through common macro-rules with similar dominant control behavior.

To clarify how the six representative rules cover the complete reduced 3x3 decision surface, Table 6 presents the rule-coverage map of the final sparse fuzzy inference system. In this table, each cell corresponds to one region of the reduced input space defined by the compressed error and change-of-error labels. The entries R1–R6 indicate which representative macro-rule is activated for each reduced decision region. This representation confirms that the proposed controller preserves full reduced input-space coverage while requiring only six explicitly defined fuzzy rules.

The reduced input space is defined as

$$\Omega = \{N, Z, P\} \times \{N, Z, P\} \quad 31$$

In this context, full reduced input-space coverage means that each cell of $\tilde{\Omega}$ is assigned to one of the representative macro-rules R1–R6. Therefore, the union of the cell sets represented by the six macro-rules covers the complete reduced 3×3 input space:

$$C^1 \cup C^2 \cup C^3 \cup C^4 \cup C^5 \cup C^6 = \Omega \quad 32$$

This definition does not imply that the original continuous input universe is redefined entirely into six crisp regions; rather, it signifies that no cell in the reduced linguistic decision matrix is left without a corresponding macro-rule. The reduction process from nine to six was carried out based on three criteria. First, each reduced cell must continue to be covered by a representative macro-rule. Second, cells assigned to a common macro-rule must either share the same dominant output polarity or belong to the same compensation-oriented control class. Thirdly, the grouping process must preserve the coarse negative, zero-action and positive control arrangement of the reduced 3×3 decision surface. Hence, the merging step is not seen as an arbitrary rule-deletion process, but rather as a macro-region representation of the reduced decision matrix, under polarity and coverage constraints.

Ce/E	N	Z	P
N	R1	R3	R2
Z	R3	R4	R5
P	R6	R4	R6

Table 6. Rule-coverage map of the final six-rule fuzzy inference system.

- It is important to point out that the proposed macro-rule grouping does not offer a new, crisp output value for each merged region. Rather, each macro-rule retains its respective dominant output linguistic label (N, Z, or P) and the controller output is still obtained by the usual Mamdani fuzzy inference procedure followed by defuzzification by the centroid method. The merged regions simplify the rule base without altering the inference mechanism and the defuzzification process. Thus, the reduction from nine decision cells to six representative rules is achieved by merging adjacent regions that share the same dominant polarity or similar compensatory behavior. Rule R3 corresponds to the (N, Z) and (Z, N) cells since both refer to a state close to equilibrium on the negative transition side. The (Z, Z) and (P, Z) cells are similar to the (Z, Z) and (P, Z) cells represented by Rule R4, since the control action required in both regions is zero (near the equilibrium and in the zero-crossing zones on the positive side). Rule R6 divides the positive side into the (P, N) and the (P, P) areas. Thus, the proposed controller reduces the number of explicitly evaluated fuzzy rules while maintaining the negative, equilibrium, and positive control regions required for speed regulation. The proposed reduction of rules from 9 to 6 is not an arbitrary compression of the rule base. The unioned regions were chosen according to neighborhood continuity and dominant control polarity. This allows to represent neighboring regions presenting similar control actions with a single macro-rule. This approach reduces the number of explicitly evaluated rules and preserves the continuity of the decision surface.

5. Result and Discussion

In the present study, the conventional speed-loop controller is replaced by fuzzy logic controllers with different rule-base sizes, while the underlying induction motor model and FOC structure are kept

unchanged. To ensure a objective evaluation, the base and reduced rule controllers are tested under identical operating conditions, motor parameters, and reference trajectories. Thus, the only varying factor among the compared cases is the fuzzy rule-base complexity.

The primary controlled comparison in this study is conducted among the original 25-rule FLC, the reduced 9-rule FLC, and the reduced 6-rule SFLC. These three controllers were constructed using the same design framework and were evaluated with identical membership-function shapes, input and output universes of discourse, scaling factors, Mamdani inference settings, aggregation method, centroid-defuzzification procedure, sampling conditions, induction-motor parameters, FOC architecture, and reference-speed profile. Therefore, the rule-base structure is the only intentionally varied controller-design factor, allowing the influence of the proposed reduction procedure to be evaluated directly. The PI controller is included only as a conventional reference benchmark and is not presented as a complexity-matched fuzzy-control competitor. Likewise, numerical superiority over previously published fuzzy controllers is not claimed because differences in machine ratings, operating profiles, membership functions, scaling factors, and implementation environments would prevent a controlled comparison. The reported conclusions are therefore based primarily on the internally controlled comparison among the three fuzzy configurations.

In addition to the conventional transient response indicators, the present study also evaluates controller performance through integral error metrics. This additional assessment is important because overshoot, rise time, and settling time primarily describe local dynamic behavior around reference transitions, whereas IAE, ITAE, and ISE provide a broader measure of cumulative tracking performance over the full operating interval. Accordingly, the effectiveness of the reduced-rule controllers is examined from both transient and accumulated-error perspectives.

The effectiveness of the proposed zone-reduction method was first evaluated by comparing the speed responses of the baseline 25-rule FLC, the reduced 9-rule FLC, and the reduced 6-rule simplified fuzzy logic controller (SFLC) under the same operating conditions.

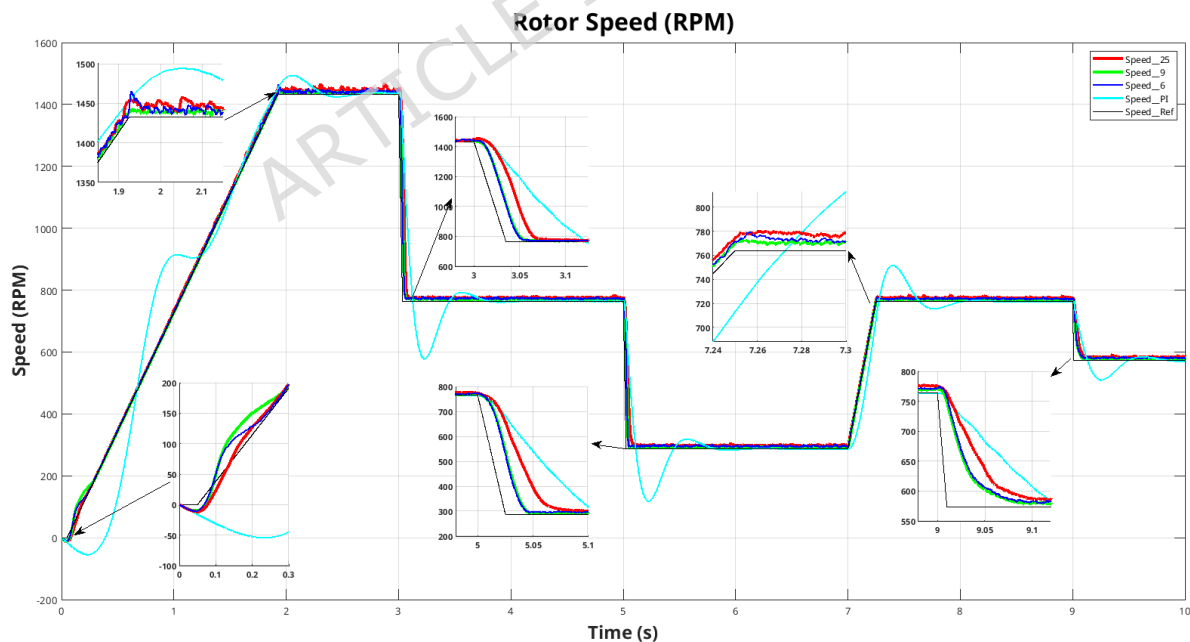


Figure 3. Speed responses of the compared controllers under the reference-speed profile.

Figure 3 shows the rotor-speed trajectories of the four controllers for the applied reference-speed profile. As observed from figure 3, all controllers were able to track the reference profile successfully throughout

the full operating interval, including the acceleration, deceleration, and intermediate speed-transition regions.

Controller	Settling Time (Ts(s))	Rising Time (Tr(s))	Overshoot (%)
25-rule FLC	1.8912	1.5047	1.2348
9-rule FLC	1.8876	1.5497	0.5051
6-rule SFLC	1.8818	1.5030	1.6714
PI Controller	2.2356	1.1888	4.7525

Table 7. Transient-response performance comparison of the considered controllers.

A deeper look at the transient region around the initial high-speed reference shows that the three controllers behave similarly. Overshoot was minimized to 1.23% with the 25-rule controller and 0.5051% with the 9-rule controller. This is essential because it indicates that rule-base reduction can increase damping and smooth transient performance. The 6-rule SFLC, however more aggressively simplified, had an adequate transient response with 1.6714% overrun. Although greater than the 25-rule and 9-rule controllers, this value is still within an acceptable range for drive applications ($\pm 2\%$). The baseline 25-rule controller settled in 1.8912 s. After decreasing the rule base to 9, the settling time dropped to 1.8876 s, nearly negligible. The settling time improved to 1.8818 s when the rule base was lowered to 6. Rise time was 1.5047 s for the basic 25-rule controller and 1.5497 s for the 9-rule controller. The 6-rule SFLC has a rise time of 1.5030 s, almost identical to the original controller.

Overall, the transient-response results indicate that rule-base reduction does not compromise the essential speed-regulation capability of the controller. Among the reduced configurations, the 9-rule FLC offers the most favorable transient trade-off, whereas the 6-rule SFLC remains a viable low-complexity alternative for computationally constrained implementations.

Controller	IAE	ITAE	ISE
25-rule FLC	1429.9495	921.1607	1.411968e6
9-rule FLC	1415.9598	912.4543	1.383354e6
6-rule SFLC	1418.1475	913.2742	1.388775e6
PI Controller	1445.5617	895.8951	1.5346 $\times 10^6$

Table 8. Integral error performance indices of the compared controllers.

Compared with the baseline 25-rule FLC (IAE = 1429.9495, ITAE = 921.1607, ISE = 1,411,968), the 9-rule FLC achieved the lowest IAE, ITAE, and ISE values among the fuzzy-controller configurations (IAE = 1415.9598, ITAE = 912.4543, ISE = 1,383,354), indicating that the first reduction stage improved tracking performance while reducing rule-base complexity. While the PI controller yielded a lower ITAE value, it exhibited a significantly higher overshoot and poorer settling performance, indicating a less favorable transient response. The 6-rule SFLC also preserved a comparable error level (IAE = 1418.1475, ITAE = 913.2742, ISE = 1,388,775), with values slightly higher than those of the 9-rule FLC but still better than the original 25-rule structure. These results confirm that the reduced controllers maintain the essential control behavior of the full rule base despite substantial simplification.

Controller	Run Time 1	Run Time 2	Run Time 3	Run Time 4	Run Time 5	Min	Max	Relative Reduction (%)
25-rule FLC	14.529	17.093	16.04	13.349	13.317	13.317	17.093	0%
9-rule FLC	10.566	10.493	10.452	10.602	9.829	9.829	10.602	30.12%
6-rule SFLC	6.823	6.172	5.999	6.143	6.254	5.999	6.823	57.77%

Table 9. Statistical comparison of profiler-based execution times in seconds for fuzzy controllers.

The execution-time comparison in Table 9 was obtained from MATLAB profiler-based software measurements under identical simulation conditions. These values represent relative total simulation execution times for the investigated controller configurations and should not be interpreted as embedded per-step inference latency or real-time hardware execution time. Therefore, the reported 57.77% reduction indicates a relative reduction in MATLAB/Simulink profiler execution time, rather than a verified microsecond-level saving on a DSP, FPGA, or microcontroller platform. Computational efficiency was evaluated through repeated call-time measurements, as summarized in Table 9. Relative reduction values were calculated with respect to the mean execution time. The 25-rule FLC produced the highest mean execution time (14.87), which decreased to 10.39 for the 9-rule FLC and to 6.28 for the 6-rule SFLC. These results indicate a relative reduction in MATLAB/Simulink execution time under the tested simulation conditions, with the 6-rule controller requiring the lowest profiler-based execution time among the investigated fuzzy-controller configurations.

The execution-time results show that the computational gain is not linear with respect to rule-count reduction. Even after the first simplification stage, the inference engine remains strongly sensitive to the number of active rules, which highlights controller compactness as a relevant design parameter for future embedded fuzzy-drive implementation.

In addition to the time-domain performance indices, an FCM-based structural validation procedure was employed to examine whether the reduced controllers preserve the dominant decision-region organization of the original fuzzy decision surface. In this analysis, FCM was not used for controller design or tuning. Instead, it was used as an external unsupervised validation tool to quantify the structural similarity between the original and reduced fuzzy decision maps.

First, the fuzzy decision surface of the baseline 25-rule controller was sampled over a uniformly distributed grid in the normalized error–change-of-error plane. For each grid point, the controller output was computed using the corresponding fuzzy inference system. Thus, each sampled point was represented by the feature vector

$$x_i = [e_i, ce_i, u_i]^T \quad (3326)$$

where e_i , ce_i , and u_i denote the normalized speed error, change in speed error, and fuzzy controller output, respectively. The sampled decision-surface data were then clustered using fuzzy c-means for different values of K , where K represents the number of structural regions used to approximate the original decision surface. The FCM objective function is defined as

$$J_m = \sum_{i=1}^N \sum_{k=1}^K \mu_{ki}^m \|x_i - v_k\|^2 \quad (3427)$$

where N is the number of sampled points, K is the number of clusters, μ_{ki} is the membership degree of sample i to cluster k , m is the fuzziness coefficient, and v_k is the k -th cluster center. After convergence,

hard assignment was performed by assigning each sample to the cluster with the maximum membership degree:

$$c_i = \arg \max_k (\mu_{ki}) \quad (285)$$

The reconstructed output surface was then obtained from the output component of the assigned cluster center. The structural similarity between the original fuzzy decision surface and the FCM-reconstructed surface was quantified using the coefficient of determination:

$$R^2 = 1 - \frac{\sum_{i=1}^N (u_i - \hat{u}_i)^2}{\sum_{i=1}^N (u_i - \bar{u})^2} \quad (296)$$

where u_i is the original fuzzy output, $\hat{u}_i$ is the reconstructed output obtained from the hard-assigned FCM region, and $\bar{u}$ is the mean of the original fuzzy output samples. Higher R^2 values indicate that the selected number of structural regions can more accurately reproduce the original fuzzy decision surface. Therefore, the FCM-based validation provides an independent measure of how much of the original decision-region topology is retained after rule-base reduction.

Rules K	25	12	9	6	5	4
R^2	0.9159	0.8789	0.8593	0.8159	0.7148	0.6568

Table 10. FCM-Based K-Region Validation Results.

As shown in Table 10, the R^2 value decreases as the number of FCM regions is reduced. The 9-region and 6-region representations yield R^2 values of 0.8593 and 0.8159, respectively, indicating that the reduced surfaces retain a measurable degree of structural similarity to the sampled original decision surface. However, these R^2 values should be interpreted only as auxiliary structural indicators. They do not constitute proof of closed-loop stability, robustness, disturbance rejection capability, or embedded implementation reliability. In this study, the FCM-based analysis is used only to complement the transient-response, integral-error, and profiler-based execution-time results by providing an additional view of decision-surface consistency.

When the transient-response results, integral error indices, computational timing measurements, and FCM-based structural validation are considered together, a clear trade-off emerges between control dependency and implementation efficiency. The 9-rule FLC provides the best overall performance–complexity compromise by preserving the dominant dynamic behavior of the baseline controller while improving several tracking metrics and reducing execution time. The 6-rule SFLC, in contrast, provides the highest level of computational simplification and remains attractive for real-time implementations in which execution efficiency is prioritized over maximum structural dependencies.

An important implication of the results is that a larger rule base does not necessarily yield superior control performance. The fact that the 9-rule controller outperformed the original 25-rule structure in overshoot and integral error indices suggests that the proposed reduction process removes local redundancy while preserving the dominant control tendencies of the decision surface. As a result, the diminishment of regulations in this research is perceived not merely as a strategy to mitigate managerial risk, but is also employed as an inherent modification.

The suggested methodology must diverge from conventional fuzzy control design techniques. Adaptive and self-regulating fuzzy controllers are typically engineered to enhance computing efficiency during execution, whereas optimization techniques employ search algorithms to modify user performance, scaling parameters, or weighting criteria. Fuzzy-neural and evolving fuzzy controllers, conversely, integrate learning processes and data-driven adjustments. The techniques outlined in this research do not

encompass online alterations, parameter revisions, or data constraints. He concentrated on refining Mamdani's conventional concepts while preserving his distinctive approach to decision-making in comparative analysis. Quantitative assessments with earlier published instruments that comprehend basic principles may lack precision due to variations in induction motor setups, device arrangements, operational parameters, and testing environments. Consequently, every category of motors examined in this study is modeled utilizing an identical array of induction motor specifications, FOC architecture, operational conditions, and reference paths.

This establishes a robust foundation for assessing the efficacy of rule reduction. The suggested technique distinguishes itself from earlier simplified rule methodologies by integrating FCM-based structural dependability with a structural zone-oriented reduction approach while taking into account both control efficacy and computing economy. Before hardware implementation or safety-critical deployment, formal stability, constraint, and robustness analyses provide mathematical guarantees for closed-loop convergence, finite system signals, disturbance rejection, and operation under parameter and model uncertainties in controller design [29]. The current research investigates an additional, pertinent inquiry: if the concept of a fixed-momentum fuzzy speed controller may be streamlined without compromising its demonstrated efficacy under identical circumstances.

To isolate the influence of rule-base restructuring, the 25-rule, 9-rule, and 6-rule controllers were evaluated using the same induction-motor model, FOC architecture, membership functions, scaling factors, inference mechanism, sampling conditions, and reference-speed trajectory. Within this controlled framework, the reduced controllers maintained comparable transient behavior and cumulative tracking performance, while providing lower MATLAB profiler execution times and measurable decision-surface similarity. Therefore, the reported results provide consistent empirical support for the proposed zone-reduction strategy within the investigated operating scenario, although they are not presented as universal formal guarantees. Lyapunov-based stability analysis, boundedness conditions, robustness margins under parameter and load variations, and hardware-oriented validation constitute complementary extensions that will broaden the applicability of the present contribution.

Overall, the results support the proposed reduction approach from dynamic, profiler-based computational, and structural similarity perspectives within the investigated simulation scenario. This multi-layer validation strategy shows that the simplified controllers preserve the essential behavior of the original fuzzy system while providing substantial reductions in implementation, thereby supporting the usage of zone-reduced fuzzy control in FOC-based induction motor drives.

6. Conclusion

This paper proposed a systematic zone-reduction approach for simplifying the fuzzy logic speed controller used in the field-oriented control of an induction motor drive. Starting from a baseline 25-rule fuzzy controller, the rule base was reduced in two stages: first to a 9-rule fuzzy controller through dominant-polarity compression and then to a 6-rule structure through macro zone merging. The reduction strategy was designed to preserve the dominant negative, equilibrium, and positive decision regions of the original fuzzy surface while decreasing the number of explicitly evaluated rules.

This study's key findings are:

- A systematic two-stage zone-reduction approach decreases the fuzzy rule base from 25 to 9 and then to 6 rules while keeping dominant decision-making behavior.
- The basic devices are assessed using accuracy results, error rate, execution time measurement, and FCM-based structural verification.
- The suggested approach demonstrates that the controller behavior and the solution space's form can be used to regulate the fuzzy controller problem
- The findings indicate that the suggested approach enhances both control efficacy and computational economy. The 9-rule fuzzy controller (FLC) achieved the optimal equilibrium between efficacy and intricacy. The overshoot diminished from 1.2348% to

0.5051% in relation to the 25-way FLC, and the fuzzy controller setup exhibited the minimal IAE, ITAE, and ISE metrics. The 6-way fuzzy controller (SFLC) diminished the execution duration by 57.77% while preserving dynamic efficacy, yielding the greatest computational advantage.

- Structural validation using FCM confirmed that the proposed reduction approach does not result in arbitrary rule deletion. The 9- and 6-region representations retained structural similarity to the original fuzzy decision surface, with R^2 values of 0.8593 and 0.8159, respectively. These results demonstrate that the reduced controllers preserve the dominant organization of the decision regions of the original rule base while simplifying implementation.

Overall, the study shows that rule-base simplification can reduce the structural complexity of a fixed Mamdani fuzzy speed controller for the investigated FOC-based induction motor simulation case. The 9-rule controller provided the most favorable balance between speed-tracking performance and rule-base compactness, whereas the 6-rule controller achieved the largest relative reduction in MATLAB profiler execution time. The FCM-based analysis provided an auxiliary indication of decision-surface similarity, but it should not be interpreted as a formal stability or robustness guarantee. Since the present study is limited to simulation-based validation, future work will focus on robustness analysis under parameter and load variations, per-step inference-time measurements on embedded processors, processor-in-the-loop or hardware-in-the-loop testing, and experimental validation on a motor-drive platform.

References

- [1] Saidur, R., Mekhilef, S., Ali, M., Safari, A., & Mohammed, H. (2012). Applications of variable speed drive (VSD) in electrical motors energy savings. *Renewable and Sustainable Energy Reviews*. <https://doi.org/10.1016/j.rser.2011.08.020>
- [2] Humza, M., & Kim, B. (2016). Design of an induction motor by pre-determination of equivalent circuit parameters. *International Journal of Applied Electromagnetics and Mechanics*. <https://doi.org/10.3233/JAE-162116>.
- [3] Finch, J. W., & Giaouris, D. (2008). Controlled AC electrical drives. *IEEE Transactions on Industrial Electronics*, 55(2), 481–491. <https://doi.org/10.1109/TIE.2007.911209>.
- [4] Bose, B. K. (2009). Power electronics and motor drives-Recent progress and perspective. *IEEE Transactions on Industrial Electronics*, 56(2), 581–588. <https://doi.org/10.1109/TIE.2008.2002726>.
- [5] El Ouanjli, N., Derouich, A., El Ghzizal, A. et al. Modern improvement techniques of direct torque control for induction motor drives - a review. *Prot Control Mod Power Syst* 4, 11 (2019). <https://doi.org/10.1186/s41601-019-0125-5>.
- [6] Kurt, H.; Kiyak, I. Artificial intelligence based outdoor lighting system control design for smart cities. *Light Eng*. 2021, 29, 110–122, <https://doi.org/10.33383/2020-044>.
- [7] Precup, R.-E., & Hellendoorn, H. (2011). A survey on industrial applications of fuzzy control. *Computers in Industry*, 62(3), 213–226. <https://doi.org/10.1016/j.compind.2010.10.001>.
- [8] Jantzen, J. (2007). *Foundations of fuzzy control*. Wiley.
- [9] Homjan, J., Santiprapan, P., & Panpean, C. (2025). Fuzzy Logic Controller Design Approach for Speed Control in Separately Excited DC Motors. *IEEJ Journal of Industry Applications*. <https://doi.org/10.1541/ieejia.25000593>.

- [10] Mendoza-Mondragon, F., Hernandez-Guzman, V. M., & Rodriguez-Resendiz, J. (2018). Robust Speed Control of Permanent Magnet Synchronous Motors Using Two-Degrees-of-Freedom Control. *IEEE Transactions on Industrial Electronics*. <https://doi.org/10.1109/TIE.2017.2786203>.
- [11] El-Barbary, Z. (2012). Fuzzy logic based controller for five-phase induction motor drive system. *Alexandria Engineering Journal*. <https://doi.org/10.1016/j.aej.2012.10.005>.
- [12] Mannan, M. A., Islam, A., Uddin, M. N., Hassan, M. K., Murata, T., & Tamura, J. (2012). Fuzzy-Logic Based Speed Control of Induction Motor Considering Core Loss into Account. *Intelligent Control and Automation*. <https://doi.org/10.4236/ica.2012.33026>.
- [13] Zeb, K., Ali, Z., Saleem, K. et al. Indirect field-oriented control of induction motor drive based on adaptive fuzzy logic controller. *Electr Eng* 99, 803–815 (2017). <https://doi.org/10.1007/s00202-016-0447-5>.
- [14] Talib, M., Ibrahim, Z., Rahim, N. A., Zulhani, R., Nordin, N., Farah, N., & Razali, A. (2020). An improved simplified rules Fuzzy Logic Speed Controller method applied for induction motor drive. *ISA Transactions*. <https://doi.org/10.1016/j.isatra.2020.05.040>.
- [15] Gacto, M. J., Alcalá, R., & Herrera, F. (2011). Interpretability of linguistic fuzzy rule-based systems: An overview of interpretability measures. *Information Sciences*, 181(20), 4340–4360. [10.1016/j.ins.2011.02.021](https://doi.org/10.1016/j.ins.2011.02.021).
- [16] Chan, T. F., & Shi, K. (2011). *Applied intelligent control of induction motor drives*. Wiley-IEEE Press. <https://doi.org/10.1002/9780470825587>.
- [17] Ali, J. A., Hannan, M. A., Mohamed, A., & Abdolrasol, M. G. (2016). Fuzzy logic speed controller optimization approach for induction motor drive using backtracking search algorithm. *Measurement*. <https://doi.org/10.1016/j.measurement.2015.09.038>.
- [18] Bachache, N. K., & Wen, J. (2012). Particle Swarm Optimize Fuzzy Logic Memberships of AC-Drive. *Lecture Notes in Computer Science*. https://doi.org/10.1007/978-3-642-30976-2_56.
- [19] Farah, N., Talib, M. H. N., Ibrahim, Z. B., Abdullah, Q., Aydoğdu, Ö., Lazi, J. M., & Isa, Z. (2021). Fuzzy membership functions tuning for speed controller of induction motor drive: performance improvement. *Indonesian Journal of Electrical Engineering and Computer Science*. <https://doi.org/10.11591/ijeecs.v23.i3.pp1258-1270>.
- [20] Mugheri, N. H., & Keerio, M. U. (2021). An Optimal Fuzzy Logic-based PI Controller for the Speed Control of an Induction Motor using the V/F Method. *Engineering, Technology & Applied Science Research*. <https://doi.org/10.48084/etasr.4255>.
- [21] Fazzolari, M., Alcalá, R., Nojima, Y., Ishibuchi, H., & Herrera, F. (2013). A review of the application of multiobjective evolutionary fuzzy systems: Current status and further directions. *IEEE Transactions on Fuzzy Systems*, 21(1), 45–65. <https://doi.org/10.1109/TFUZZ.2012.2201338>.
- [22] Yan, S., Hu, X., & Park, J. H. (2026). Deep-Q-learning-based proportional-integral event-triggered fuzzy output control of variable-speed AGVs via proportional-integral observer. *IEEE Transactions on Fuzzy Systems*, 34(6), 1833–1843. <https://doi.org/10.1109/TFUZZ.2026.3678977>.
- [23] Yan, S., & Yang, X. (2026). Neural network-based attack-compensation control of T–S fuzzy systems against actuator attacks with improved dynamic memory-event-triggered scheme. *IEEE Transactions on Fuzzy Systems*, 34(1), 89–100. <https://doi.org/10.1109/TFUZZ.2025.3625147>.
- [24] Yan, S., Gu, Z., Park, J. H., & Xie, X. (2023). Synchronization of delayed fuzzy neural networks with probabilistic communication delay and its application to image encryption. *IEEE Transactions on Fuzzy Systems*, 31(3), 930–940. <https://doi.org/10.1109/TFUZZ.2022.3193757>.

- [25] Yan, S., Ding, L., & Cai, Y. (2025). Memory-based attack-tolerant T–S fuzzy control of networked artificial pancreas system subject to false data injection attacks. *Fuzzy Sets and Systems*, 518, 109486. <https://doi.org/10.1016/j.fss.2025.109486>.
- [26] Yan, S., Gu, Z., Park, J. H., & Xie, X. (2023). Sampled memory-event-triggered fuzzy load frequency control for wind power systems subject to outliers and transmission delays. *IEEE Transactions on Cybernetics*, 53(6), 4043–4053. <https://doi.org/10.1109/TCYB.2022.3224386>.
- [27] Precup, R.-E., Haidegger, T., Preitl, S., Benyó, Z., Paul, A. S., & Kovács, L. (2012). Fuzzy control solution for telesurgical applications. *Applied and Computational Mathematics*, 11(3), 378–397. <http://www.acmij.az/view.php?lang=az&menu=cjournal&id=289>.
- [28] Hu, X., Xia, Y., Lendek, Z., Cao, J., & Precup, R.-E. (2025). A novel dynamic prescribed performance fuzzy-neural backstepping control for PMSM under step load. *Neural Networks*, 190, 107627. <https://doi.org/10.1016/j.neunet.2025.107627>.
- [29] Andonovski, G., Leite, D. F., Precup, R.-E., Gomide, F., Pratama, M., & Škrjanc, I. (2026). Advancements in data-driven evolving fuzzy and neuro-fuzzy control: A comprehensive survey. *Applied Soft Computing*, 186, 114058. <https://doi.org/10.1016/j.asoc.2025.114058>.
- [30] Mao, B., Wu, X., Liu, H., Xu, Y., & Lü, J. (2024). Adaptive fuzzy tracking control with global prescribed-time prescribed performance for uncertain strict-feedback nonlinear systems. *IEEE Transactions on Cybernetics*, 54(9), 5217–5230. <https://doi.org/10.1109/TCYB.2024.3366177>.
- [31] Orłowska-Kowalska, T., Blaabjerg, F., & Rodríguez, J. (Eds.). (2014). *Advanced and intelligent control in power electronics and drives*. Springer. <https://doi.org/10.1007/978-3-319-03401-0>.
- [32] Workineh, T. G., Jember, Y. B., & Kassie, A. T. (2023). Evaluation of intelligent PPI controller for the performance enhancement of speed control of induction motor. *Scientific African*. <https://doi.org/10.1016/j.sciaf.2023.e01982>.
- [33] Havens, T. C., Bezdek, J. C., Leckie, C., Hall, L. O., & Palaniswami, M. (2012). Fuzzy c-means algorithms for very large data. *IEEE Transactions on Fuzzy Systems*, 20(6), 1130–1146. <https://doi.org/10.1109/TFUZZ.2012.2201485>.
- [34] Wang, H., Wang, J., & Zhu, L. (2021). A new validity function of FCM clustering algorithm based on intra-class compactness and inter-class separation. *Journal of Intelligent & Fuzzy Systems*. <https://doi.org/10.3233/JIFS-210555>.
- [35] Eustaquio, F., & Nogueira, T. (2020). Evaluating the numerical instability in fuzzy clustering validation of high-dimensional data. *Theoretical Computer Science*. <https://doi.org/10.1016/j.tcs.2019.10.039>.

Data Availability

All data analyzed during this study are included in this published article. Further information is available from the corresponding author upon reasonable request.

Code Availability

The MATLAB/Simulink models, fuzzy-rule configurations, MATLAB scripts used for post-processing, MATLAB profiler evaluation, and FCM-based decision-surface analysis will be made available to the

reviewers during the review process. Upon acceptance, the files will be provided as Supplementary Software or deposited in a public repository, subject to journal policy and institutional restrictions.

Author contributions

H.K. contributed to the conceptualization of the study, methodology development, software implementation, visualization, mathematical modelling and preparation of the original manuscript draft. A.U. contributed to the simulation studies, formal analysis and supervision of the research. P.B. contributed to data curation and validation. M.A. contributed to control parameter analysis. H.N. contributed to the manuscript revision. All authors reviewed the manuscript and approved the final version.

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Conflicts of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.